figure 14.16, showing a directed, weighted graph that models the map shown in figure 14.15.

In this context, the minimum distance between two vertices u and v is the minimum sum, across all paths from u to v, of the weights of edges in the path. If there is no such path, i.e. if there is no way to go from u to v, then the distance between them is considered to be infinite.

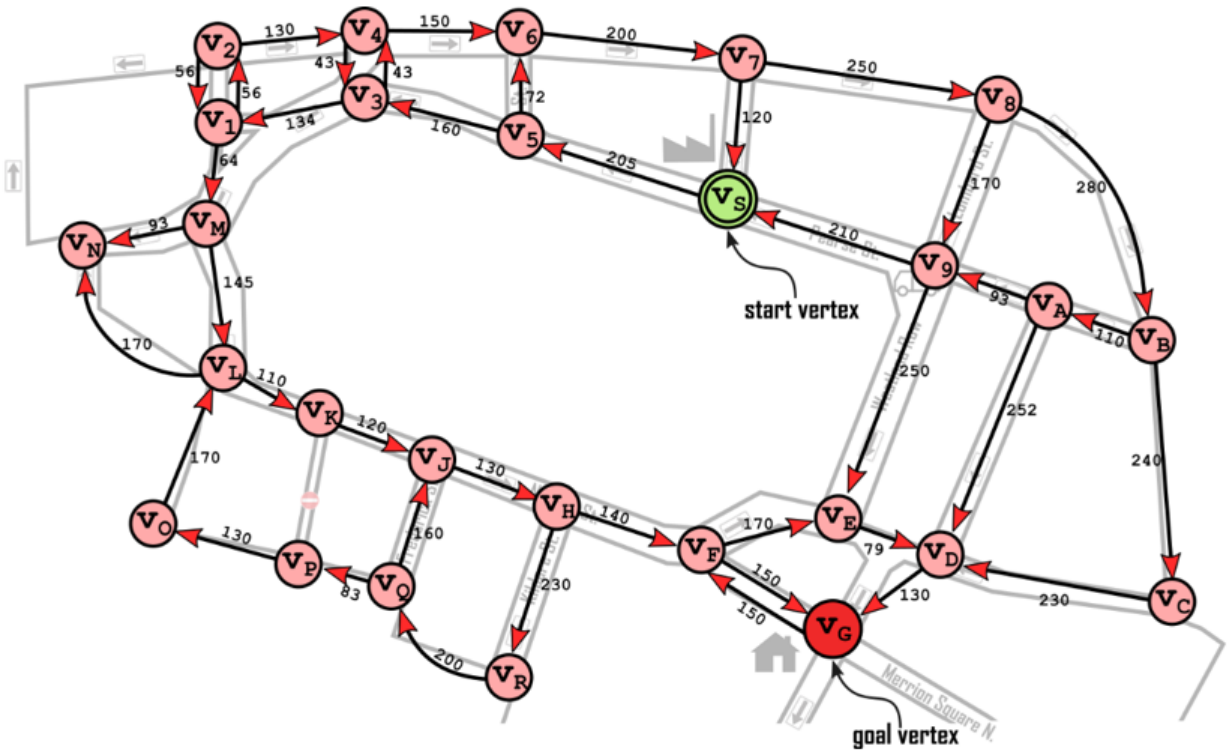


Figure 14.16 Overlaying a directed weighted graph over the map in figure 14.15. Edges' weights are the distances (in meters) between the intersections modeled by the edge's vertices.

Figure 14.17 shows, on a simpler example graph, how Dijkstra's algorithm works: it is similar to BFS, with two main differences: